

Online Appendix for
The Unintended Consequences of Raising the Standard of
Proof in Criminal Sentencing

Spencer Cooper^{*} & Andy Ye Yuan[†]

^{*} Assistant Professor of Economics, University of Connecticut

[†] Assistant Professor of Law, University of Florida Levin College of Law.

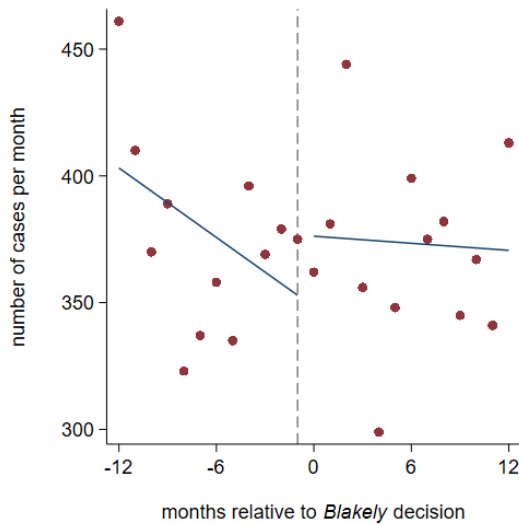
Table of Contents

Appendix A: Additional Figures	3
Figure A.1 Smoothness of Other Characteristics	3
Figure A.2 Propensity Score Distribution.....	4
Figure A.3 <i>Blakely</i> Effects on Low Probability Cases	5
Figure A.4 Changes in Prosecutor Decisions	6
Figure A.5 Changes in Prosecutor Decisions – Full Sample	8
Figure A.6 Leniency Measure Correlation	10
Figure A.7 Jury Trial Cases Sentence Discontinuity – Full Sample and Residuals	11
Figure A.8 Sample Restriction at the <i>Blakely</i> Cutoff	12
Appendix B: Additional Tables.....	14
Table B1. The Effect of <i>Blakely</i> on Enhancement Rate – Event Study	14
Table B.2 Effect of Enhancement on Sentence Length – Event Study	15
Table B.3 Effect of <i>Blakely</i> on Sentence Length – Event Study	16
Table B.4 <i>Blakely</i> Effects – Kernel and Quadratic Robustness	17
Table B.5 <i>Blakely</i> Effects –Robustness Across Various Standard Errors	18
Table B.6 <i>Blakely</i> Effects for Low Probability Cases	19
Table B.7 Placebo Discontinuities at Alternative Cutoff Dates	20
Table B.8 <i>Blakely</i> Effects on Sentence Length – Jury Trial Cases Only	21
Table B.9 Heterogeneous Effects of <i>Blakely</i> by Judge Leniency	22
Table B.10 Enhancement Effects by Judge Leniency – RD Design.....	23
Table B.11 Sentence Effects by Judge Leniency – RD Design	24
Table B.12 Judicial Heterogeneity for Low Probability Cases.....	26
Table B.13 Binding Sentencing Guidelines Mechanism Analysis	25
Table B.14 Heterogeneous Effects of <i>Blakely</i> by Other Judge Characteristics.....	27
Table B.15 Sample Restrictions and the <i>Blakely</i> Discontinuity	28
Appendix C: A Simple Model of Judicial Sentencing Under Reduced Information.....	29
Appendix D: More Institutional Details	31
D.1: Sentencing Guidelines.....	31
D.2 Statutory Aggravating Factors	32
Appendix E: Data cleaning	34

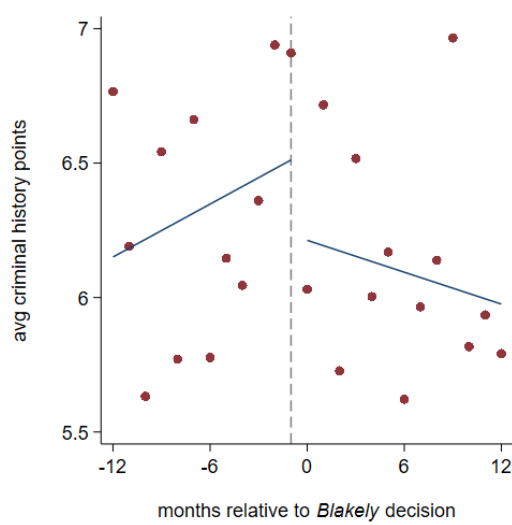
Appendix A: Additional Figures

Figure A.1 Smoothness of Other Characteristics

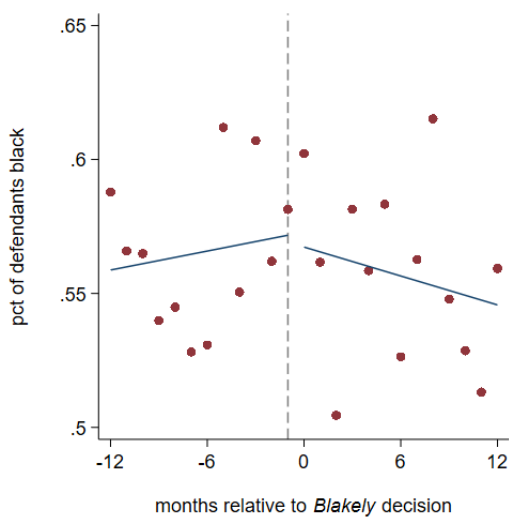
Panel (A): Case Density



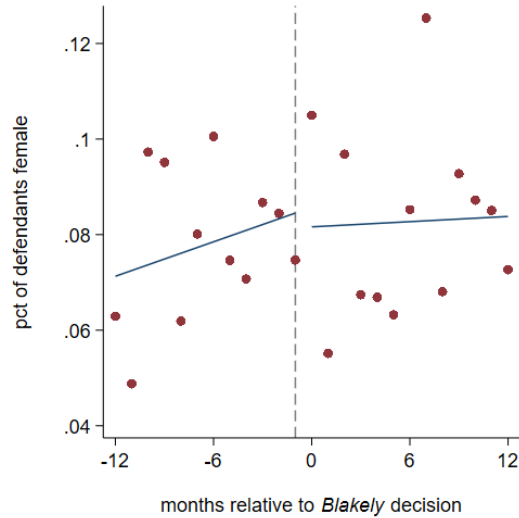
Panel (B): Criminal history points



Panel (C): Black

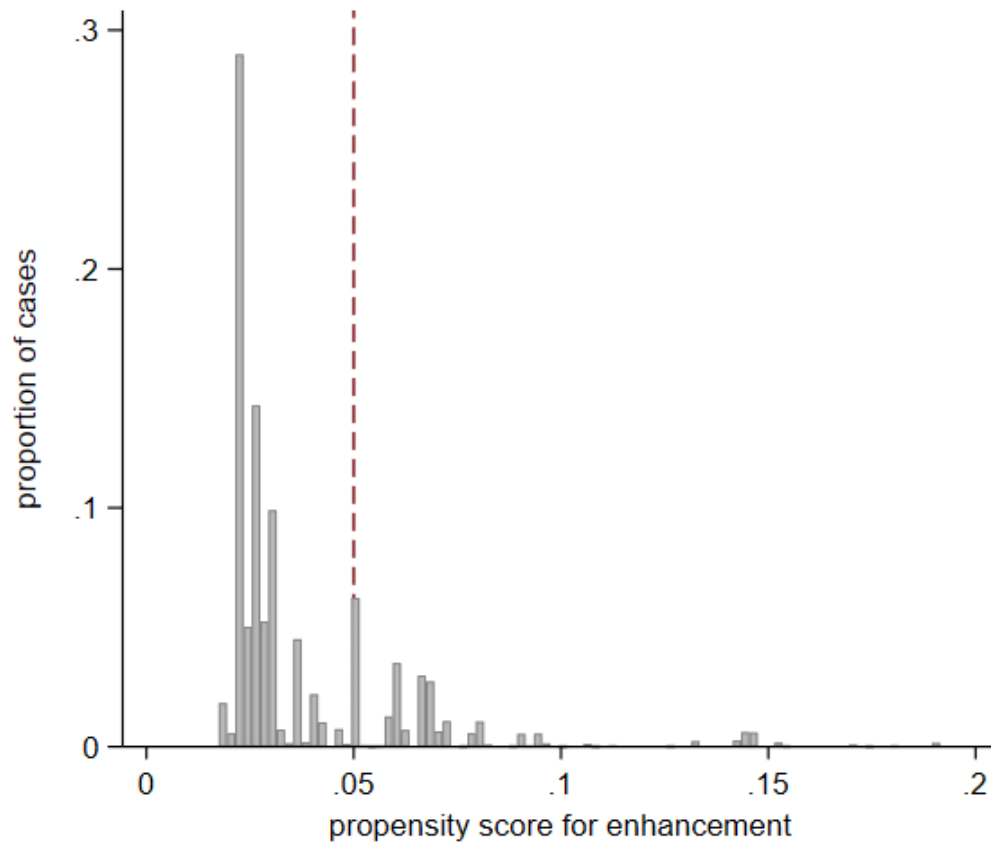


Panel (D): Female



Notes: This figure presents additional smoothness checks to ensure the sample is not changing across the timing of *Blakely*. Panel A gives the number of cases per month, while the other panels give averages of defendant characteristics. All figures represent tests for the $\text{phat} \geq 0.05$ sample.

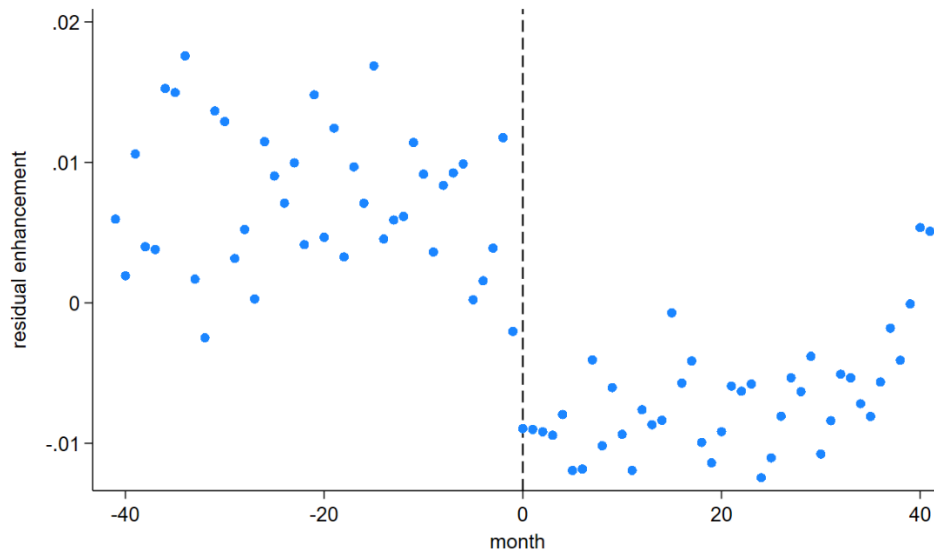
Figure A.2 Propensity Score Distribution



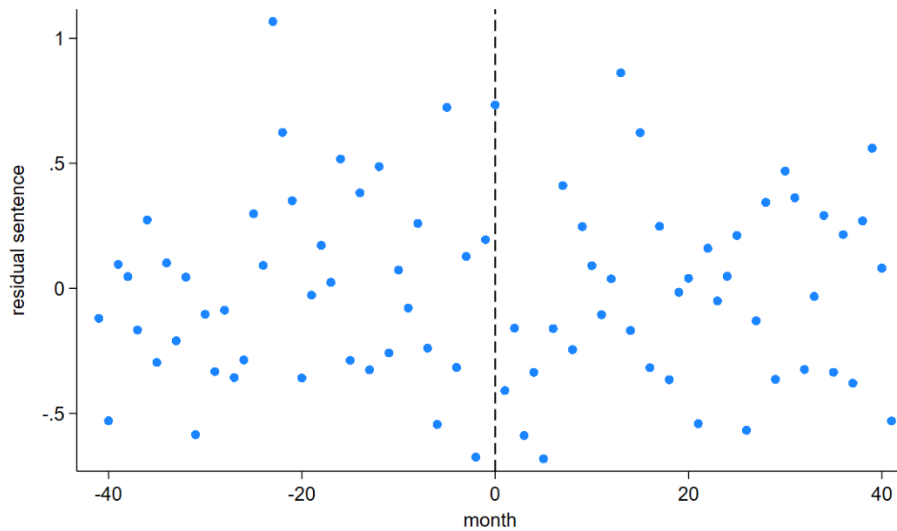
Notes: This histogram shows the distribution of cases by the estimated propensity of enhancement. The propensity score was estimated using a logit on cases decided before the timing of *Blakely*. See Section 4.2 for more details on the propensity score estimation. The dashed line at 0.05 indicates where the sample is separated for much of the analysis. In many cases we drop cases with a propensity score under 0.05. This decreases the sample size considerably, allowing for a higher proportion of cases likely to receive enhancement.

Figure A.3 *Blakely* Effects on Low Probability Cases

Panel (A): Enhancement



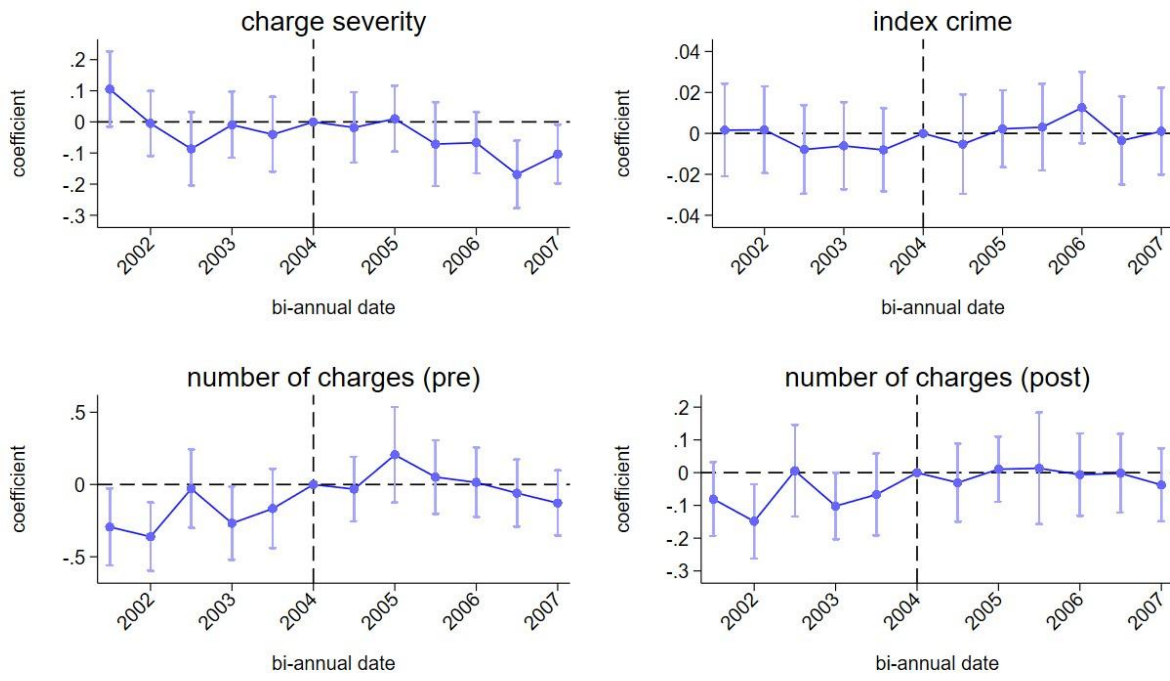
Panel (B): Sentence Length



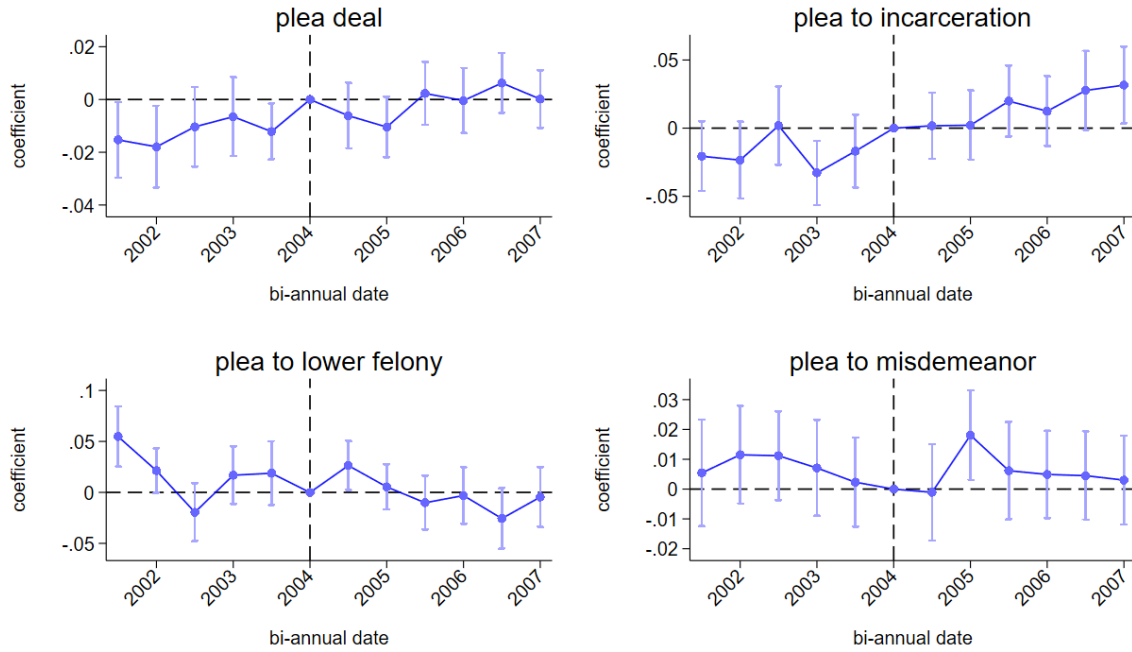
Notes: These graphs show the residual discontinuity for enhancement and sentence length at the timing of *Blakely* for those cases with a propensity score less than 0.05. Outcomes are residualized using grid fixed effects.

Figure A.4 Changes in Prosecutor Decisions

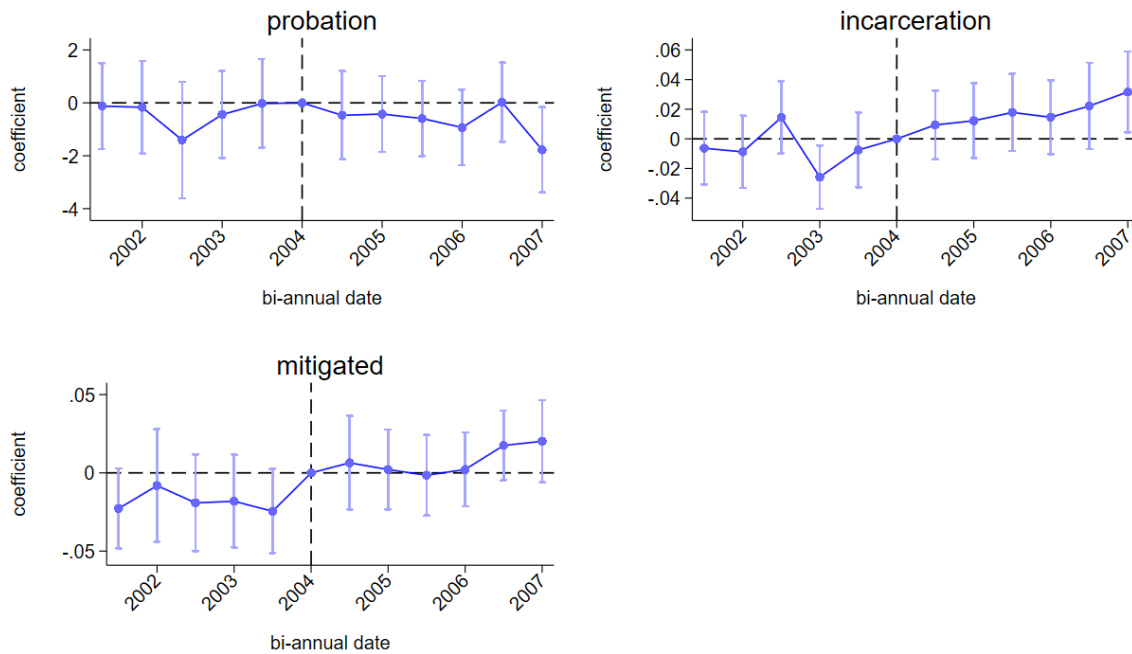
Panel (A): Charging outcomes



Panel (B): Plea bargain outcomes



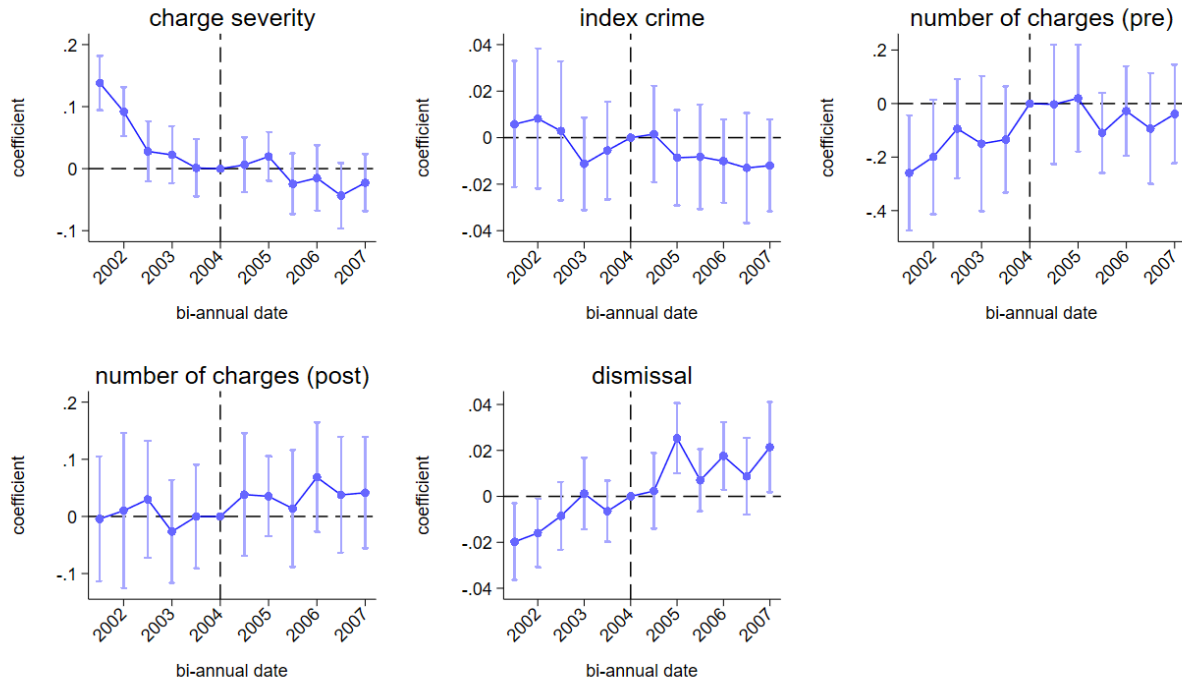
Panel (C): Sentencing outcomes



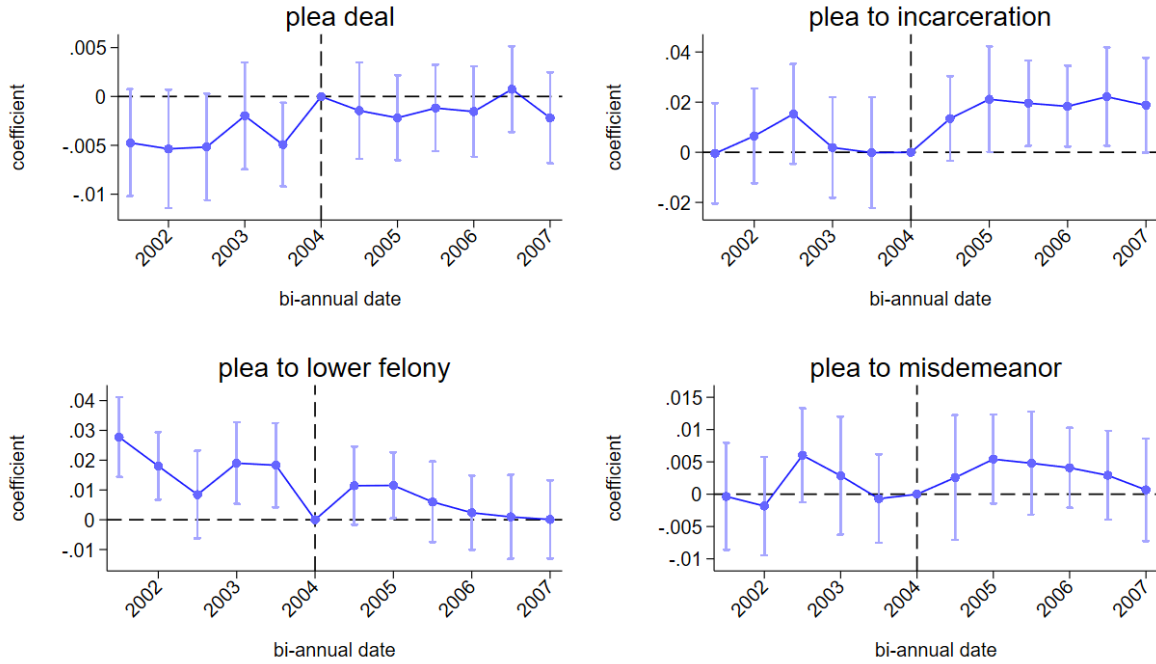
Notes: This graph gives the event study plots for outcomes related to prosecutor choice. Each regression includes grid fixed effects. The sample restricted to cases with propensity scores above 0.05 is used throughout.

Figure A.5 Changes in Prosecutor Decisions – Full Sample

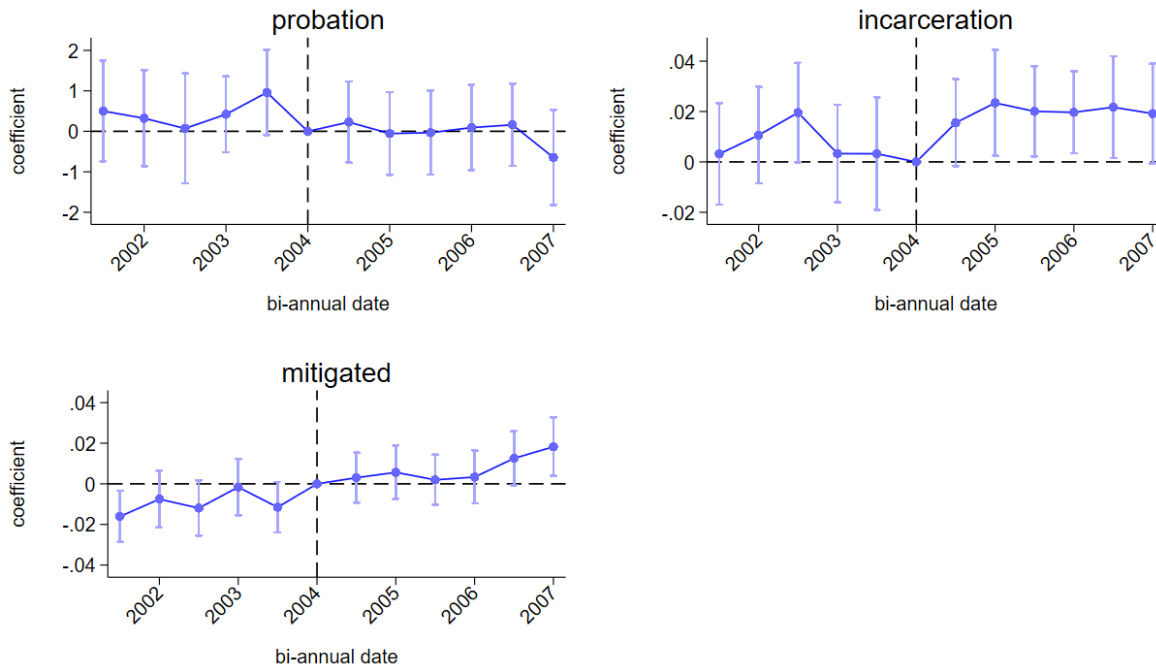
Panel (A): Charging outcomes



Panel (B): Plea bargain outcomes

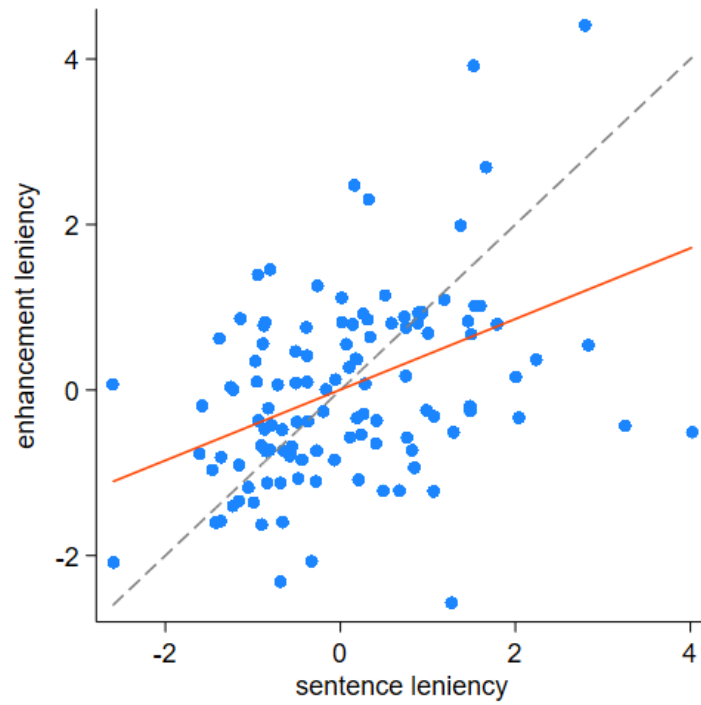


Panel (C): Sentencing outcomes



Notes: This figure reports event study analysis for each of the prosecutor outcomes as in Figure A4, but now includes the full sample of all cases.

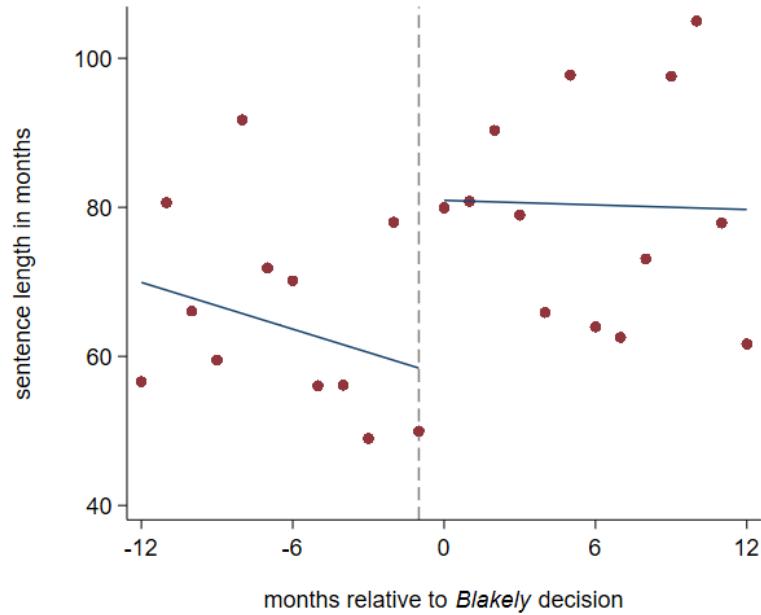
Figure A.6 Leniency Measure Correlation



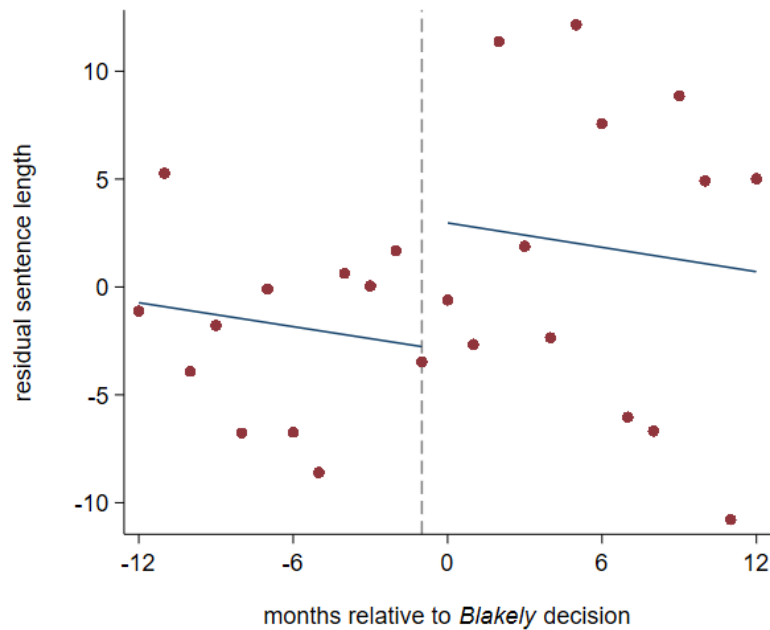
Notes: We plot the two leniency measures against each other to see to what degree judges are lenient across sentence length and enhancement. The dashed 45-degree line gives reference for a judge that is equally lenient/harsh by sentence and enhancement measures. The orange line gives the actual linear correlation between these two measures. Each leniency measure is standardized to have a mean of 0 and a standard deviation of 1. The figure is truncated to exclude one outlier judge for readability.

Figure A.7 Jury Trial Cases Sentence Discontinuity – Full Sample and Residuals

Panel (A): Full Sample



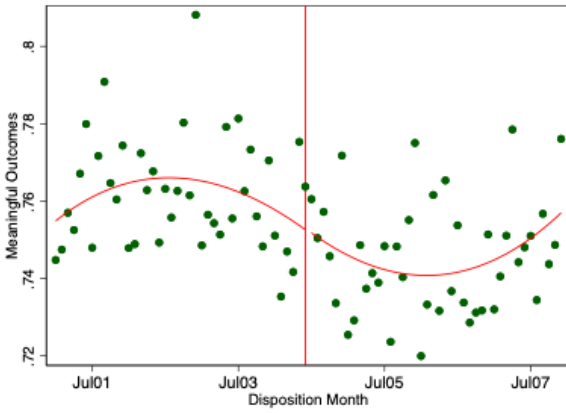
Panel (B): Residuals



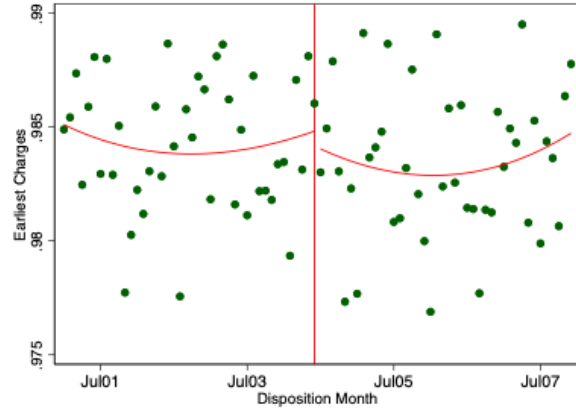
Notes: These figures show the discontinuity in sentence length at the time of *Blakely* for cases resolved at jury trial. Panel A contains all trial cases within the 12-month bandwidth of *Blakely* while Panel B gives results for the phat restricted sample and plots residuals rather than raw sentence lengths.

Figure A.8 Sample Restriction at the *Blakely* Cutoff

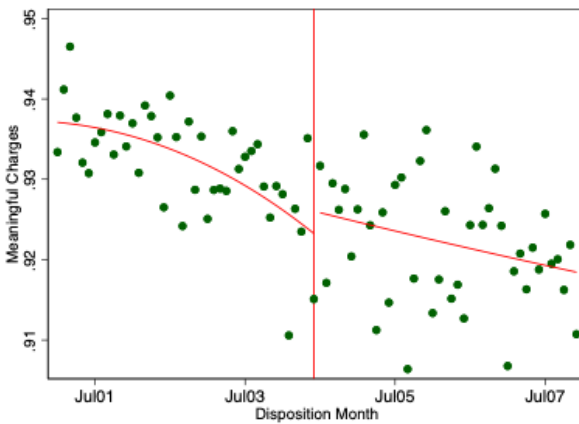
Panel (A): Meaningful Outcomes



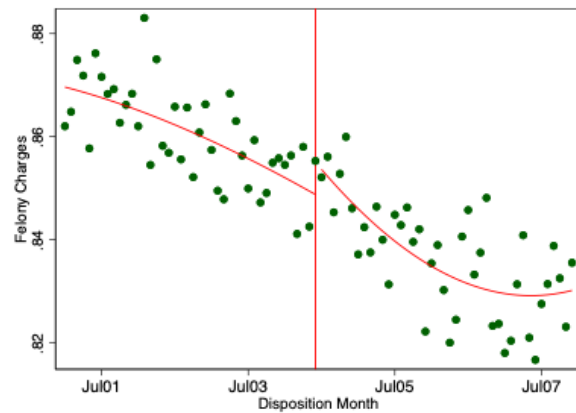
Panel (B): Earliest Charges



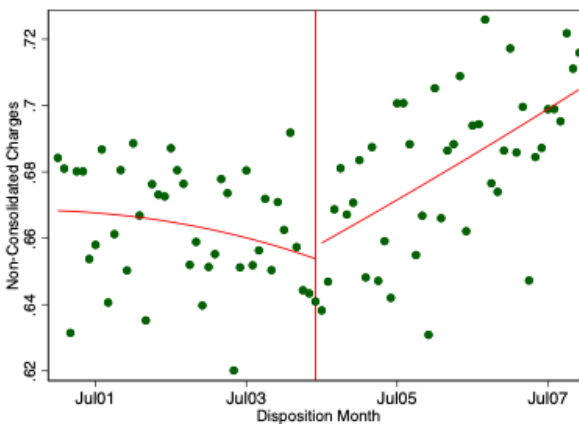
Panel (C): Meaningful charges



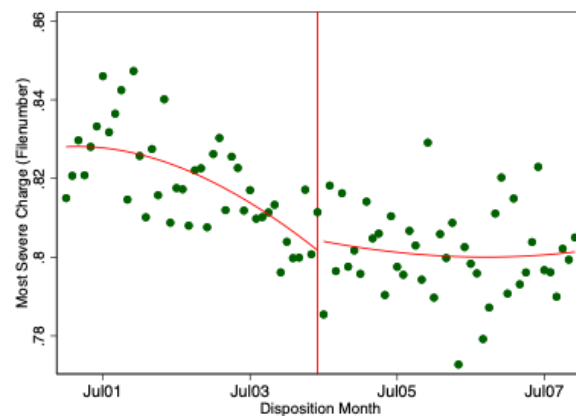
Panel (D): Felony Charges



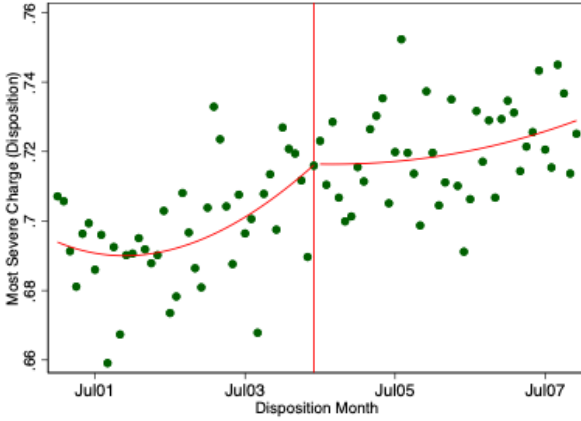
Panel (E): Non-consolidated Charges



Panel (F): Most Severe Charge 1



Panel (G): Most Severe Charge 2



Notes: The figures show the fraction of observations that survived each sample restriction and the changes in the fractions at the *Blakely* cutoff. The line in the middle is the month when *Blakely* was decided. Appendix D has detailed explanation on the meaning of each sample restriction. The curves are fitted on each side of *Blakely* with a second-degree polynomial function of the time. The dataset comes from the North Carolina Administrative Office of Courts (AOC). The sample period is from 2001 to 2007.

Appendix B: Additional Tables

Table B1. The Effect of *Blakely* on Enhancement Rate – Event Study

	(1)	(2)	(3)	(4)
Blakely	-0.0282*** (0.00625)	-0.0282*** (0.00627)	-0.0288*** (0.00630)	-0.0292*** (0.00643)
Pre-Blakely mean	0.0721	0.0721	0.0721	0.0721
fixed effects	yes	yes	yes	yes
defendant controls	no	yes	yes	yes
counsel/crime controls	no	no	yes	yes
arrest/unemp/population	no	no	no	yes
N	27636	27636	27636	27636

Notes: The dependent variable is a binary measure for whether a case is sentenced with enhancement. Standard errors clustered at the judicial district level are presented in parentheses. Grid, county, and year fixed effects are included in each specification. Defendant controls include age, gender, race, and criminal history points. Counsel/crime controls include binaries for whether the defense type is a private attorney or a public defender and a set of specific crime type fixed effects. The specification in column 4 also controls for the county-year level arrest rate, population level, and unemployment rate. The p-hat restriction is based on the propensity score described above. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.2 Effect of Enhancement on Sentence Length – Event Study

	(1)	(2)	(3)	(4)
Aggravated	18.18*** (1.056)	18.16*** (1.056)	17.86*** (1.057)	17.85*** (1.055)
Pre-Blakely mean	39.2	39.2	39.2	39.2
fixed effects	yes	yes	yes	yes
defendant controls	no	yes	yes	yes
counsel/crime controls	no	no	yes	yes
arrest/unemp/population	no	no	no	yes
N	13177	13177	13177	13177

Notes: The dependent variable is the minimum sentence length given in months at the time of disposition. Sample only considers pre-*Blakely* observations 3 years before the timing of *Blakely*. Standard errors clustered at the judicial district level are presented in parentheses. Grid, county, and year fixed effects are included in each specification. Defendant controls include age, gender, race, and criminal history points. Counsel/crime controls include binaries for whether the defense type is a private attorney or a public defender and a set of specific crime type fixed effects. The specification in column 4 also controls for the county-year level arrest rate, population level, and unemployment rate. The p-hat restriction is based on the propensity score described above. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.3 Effect of *Blakely* on Sentence Length – Event Study

	(1)	(2)	(3)	(4)
<i>Blakely</i>	0.932 (0.550)	0.912 (0.549)	0.825 (0.566)	0.912 (0.578)
pre- <i>Blakely</i> sentence length	39.2	39.2	39.2	39.2
$\hat{\Delta}_{blakely}$ (benchmark)	-0.532	-0.531	-0.533	-0.540
t-score (benchmark)	2.66	2.63	2.40	2.51
fixed effects	yes	yes	yes	yes
defendant controls	no	yes	yes	yes
counsel/crime controls	no	no	yes	yes
arrest/unemp/population	no	no	no	yes
N	27536	27536	27536	27536

Notes: The dependent variable is the minimum sentence length given in months at the time of disposition. All specifications use the sample that includes only cases with a propensity score above 0.05 3 years before and after the timing of *Blakely*. $\hat{\Delta}_{blakely}$ gives the null hypothesis value predicted effect of *Blakely*. t-score (benchmark) gives the t-score value calculated using this benchmark value as the null hypothesis. Standard errors are clustered at the district level, presented in parentheses. Each specification includes grid fixed, county, and year fixed effects.

Table B.4 *Blakely* Effects – Kernel and Quadratic Robustness

	(1)	(2)	(3)	(4)	(5)	(6)
	enhanced	enhanced	enhanced	sentence length	sentence length	sentence length
<i>Blakely</i>	-0.0225** (0.0112)	-0.0205* (0.0109)	-0.0257* (0.0150)	1.522* (0.842)	1.457* (0.825)	1.074 (1.357)
kernel	triangle	epanech.	uniform	triangle	epanech.	uniform
polynomial order	1	1	2	1	1	2
N	8440	8440	9314	8440	8440	9314

Notes: This table presents results for the main discontinuity analysis again using several different kernels and a second order polynomial. All specifications consider *Blakely* effects using the 0.05 restricted propensity score sample and a 12-month bandwidth. Standard errors presented in parentheses are clustered at the month level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.5 *Blakely* Effects –Robustness Across Various Standard Errors

	(1)	(2)	(3)	(4)	(5)	(6)
	enhanced	enhanced	enhanced	sentence length	sentence length	sentence length
<i>Blakely</i>	-0.0208**	-0.0208*	-0.0208**	1.996***	1.996**	1.996**
	(0.0089)	-	(0.0097)	(0.6795)	-	(0.7807)
p-value	0.021	0.056	0.032	0.003	0.022	0.011
SE type	EHW	wild bootstrap	cluster on district	EHW	wild bootstrap	cluster on district
N	9314	9314	9314	9314	9314	9314

Notes: This table presents results for the main discontinuity analysis again using different standard errors for inference. Following motivation given in Kolesar & Rothe (2018), columns 1 and 4 use conventional Eicker-Huber-White standard errors. Columns 2 and 5 use wild clustered bootstrap standard errors clustered at the month level. We also check that clustering on geographic level does not matter in columns 3 and 6, which cluster at the district level. All specifications use a 12-month bandwidth, a uniform kernel, and a local linear design.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.6 *Blakely* Effects for Low Probability Cases

	(1) enhanced	(2) enhanced	(3) sentence length	(4) sentence length
<i>Blakely</i>	-0.0119*** (0.0033)	-0.0096** (0.0048)	0.0414 (0.387)	0.206 (0.231)
p-hat restriction	< 0.05	< 0.03	< 0.05	< 0.03
N	30889	23839	30722	23691

Notes: The following estimates give regression discontinuity results for cases with low propensity of enhancement. Samples in columns 1 and 3 include cases with a propensity score less than 0.05 and columns 2 and 4 include those with a score less than 0.03. All specifications use standard errors clustered at the month level, a 12-month bandwidth, a uniform kernel function, and include grid fixed effects.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.7 Placebo Discontinuities at Alternative Cutoff Dates

	(1) enhancement	(2) enhancement	(3) min. sent	(4) min. sent
<i>Blakely</i>	-0.0022 (0.0087)	0.0022 (0.0131)	0.0061 (0.829)	0.376 (0.699)
Cutoff date	July 2006	July 2002	July 2006	July 2002
N	10049	9204	10016	9165

Notes: This table presents placebo regression discontinuity-in-time estimates using alternative treatment dates.

Placebo cutoffs are set two years before and two years after *Blakely* (July 2002 and July 2006), holding the month of the year constant. All specifications use the propensity score-restricted sample (propensity score > 0.05), a 12-month bandwidth, and a uniform kernel, consistent with the preferred specification. Standard errors, reported in parentheses, are clustered at the month level.

* p < 0.1, ** p < 0.05, *** p < 0.01

Table B.8 *Blakely* Effects on Sentence Length – Jury Trial Cases Only

	(1)	(2)	(3)	(4)	(5)
<i>Blakely</i>	3.791 (4.435)	8.670** (4.368)	3.796 (4.363)	4.799 (3.839)	13.10*** (4.686)
Pre- <i>Blakely</i> mean	106.1	106.3	106.0	72.8	95.3
p-hat restriction	≥ 0.05	≥ 0.05	≥ 0.05	none	≥ 0.03
bandwidth	9	12	15	12	12
N	341	446	546	853	566

Notes: This table presents results for the main sentence length discontinuity analysis again using several different kernels and a second order polynomial. All specifications consider *Blakely* effects using the 0.05 restricted propensity score sample and a 12-month bandwidth. Standard errors presented in parentheses are clustered at the month level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.9 Heterogeneous Effects of *Blakely* by Judge Leniency

	(1) enhancement	(2) enhancement	(3) min. sent	(4) min. sent
<i>Blakely</i>	0.0044 (0.0067)	0.0027 (0.0066)	1.569** (0.664)	1.160* (0.694)
DD_50	-0.0210*** (0.0059)	-0.0219*** (0.0058)	-0.687 (0.881)	-0.784 (0.781)
DD_75	-0.0403*** (0.0056)	-0.0391*** (0.0057)	-0.246 (0.595)	-0.135 (0.608)
DD_100	-0.0707*** (0.0080)	-0.0709*** (0.0080)	-1.947*** (0.549)	-1.919*** (0.540)
Fixed effects	yes	yes	yes	yes
Additional controls	no	yes	no	yes
Pre- <i>Blakely</i> mean	0.0723	0.0723	39.20	39.20
N	27536	27536	27536	27536

Notes: This table is analogous to Table 6 in the main text, but instead uses the enhancement leniency measure rather than the sentence length one. It gives the differential effects of *Blakely* on enhancement rate and sentence length across the four judge leniency groups. The *Blakely* coefficient gives the effect for the most lenient judges, while DD_50, DD_75, and DD_100 give the difference-in-difference estimates for increasingly strict judges. All specifications include grid and year fixed effects. The additional controls include age, a male dummy, race, defense type, criminal history points, the type of crime, and annual arrest and population rates. Standard errors are clustered at the month level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.10 Enhancement Effects by Judge Leniency – RD Design

	(1)	(2)	(3)	(4)	(5)	(6)
leniency_50	0.0134 (0.0106)	0.0142 (0.00898)	0.0174** (0.00781)	0.0190 (0.0119)	0.0143 (0.00989)	0.0153* (0.00855)
leniency_75	0.0331*** (0.0109)	0.0335*** (0.00951)	0.0350*** (0.00834)	0.0344*** (0.0121)	0.0337*** (0.0104)	0.0308*** (0.00908)
leniency_100	0.0589*** (0.0122)	0.0722*** (0.0109)	0.0706*** (0.00930)	0.0655*** (0.0133)	0.0706*** (0.0115)	0.0636*** (0.00972)
<i>Blakely</i>	0.000511 (0.0.122)	0.00418 (0.0107)	-0.00395 (0.00937)	0.00322 (0.0128)	0.00533 (0.0110)	-0.00510 (0.00957)
leniency_50* <i>Blakely</i>	-0.0105 (0.0133)	-0.0127 (0.0114)	-0.0167* (0.0100)	-0.0105 (0.0144)	-0.0107 (0.0119)	-0.0110 (0.0104)
leniency_75* <i>Blakely</i>	-0.0202 (0.0138)	-0.0224* (0.0120)	-0.0290*** (0.0105)	-0.0209 (0.0147)	-0.0242* (0.0125)	-0.0279** (0.0110)
leniency_100* <i>Blakely</i>	-0.0585*** (0.0137)	-0.0698*** (0.0124)	-0.0668*** (0.0109)	-0.0674*** (0.0150)	-0.0741*** (0.0130)	-0.0659*** (0.0113)
p-hat restriction	≥ 0.05	≥ 0.05	≥ 0.05	≥ 0.05	≥ 0.05	≥ 0.05
bandwidth	9	12	15	9	12	15
demographic controls	no	no	no	yes	yes	yes
county fixed effects	no	no	no	yes	yes	yes
grid fixed effects	yes	yes	yes	yes	yes	yes
N	6987	9369	11627	6978	9356	11611

Notes: The dependent variable is a binary variable for whether the case received enhancement. The leniency variable is determined according to judge propensity for enhancement, as described in Section 6.1. Variables leniency_50, leniency_75, and leniency_100 are binary variables signifying whether the assigned judge for each case is in the 2nd, 3rd, or top quartile of harshness, respectively. Here, leniency is based on the percent of a judge's cases that receive enhancement. The bottom quartile (most lenient judges) variable is the omitted group. Demographic controls include age, sex, race, criminal history, and representation controls. Standard errors are presented in parentheses. Note that we use conventional Eicker-White following Kolesar & Rothe (2018). The p-hat restriction is based on the propensity score described above. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.11 Sentence Effects by Judge Leniency – RD Design

	(1)	(2)	(3)	(4)	(5)	(6)
leniency_50	1.076 (0.765)	2.029*** (0.651)	2.057*** (0.607)	1.189 (0.905)	2.039*** (0.740)	2.158*** (0.662)
leniency_75	1.889** (0.773)	3.081*** (0.680)	3.092*** (0.640)	2.360*** (0.952)	3.307*** (0.781)	3.042*** (0.709)
leniency_100	4.215*** (0.884)	5.844*** (0.777)	5.463*** (0.694)	4.948*** (0.992)	6.030*** (0.847)	5.562*** (0.769)
<i>Blakely</i>	1.652 (1.036)	3.748*** (0.905)	2.718*** (0.813)	2.085** (1.042)	3.839*** (0.915)	2.852*** (0.821)
leniency_50* <i>Blakely</i>	-0.244 (1.083)	-1.490 (0.918)	-1.658** (0.835)	-0.288 (1.137)	-1.048 (0.953)	-1.449* (0.853)
leniency_75* <i>Blakely</i>	-1.139 (1.137)	-2.506*** (0.970)	-2.273*** (0.903)	-1.834** (1.208)	-2.646*** (1.008)	-2.531*** (0.928)
leniency_100* <i>Blakely</i>	-1.015 (1.221)	-3.403*** (1.056)	-3.150*** (0.940)	-1.911 (1.259)	-3.694*** (1.082)	-3.328*** (0.960)
p-hat restriction	≥ 0.05	≥ 0.05	≥ 0.05	≥ 0.05	≥ 0.05	≥ 0.05
bandwidth	9	12	15	9	12	15
demographic controls	no	no	no	yes	yes	yes
county fixed effects	no	no	no	yes	yes	yes
grid fixed effects	yes	yes	yes	yes	yes	yes
N	6966	9339	11589	6957	9326	11573

Notes: The dependent variable is the minimum sentence length given in months at the time of disposition. The leniency variable is a mean of other sentence lengths assigned by the judge, as described in Section 6.1. Variables leniency_50, leniency_75, and leniency_100 are binary variables signifying whether the assigned judge for each case is in the 2nd, 3rd, or top quartile of harshness, respectively. The bottom quartile (most lenient judges) variable is the omitted group. Demographic controls include age, sex, race, criminal history, and representation controls. Standard errors are presented in parentheses. Note that we use conventional Eicker-Huber-White following Kolesar & Rothe (2018). The p-hat restriction is based on the propensity score described above.

Table B.12 Binding Sentencing Guidelines Mechanism Analysis

	(1) top guideline	(2) top guideline	(3) top guideline	(4) top guideline
sent_leniency_50	0.0080 (0.0300)	0.0004 (0.0288)	0.0052 (0.0398)	-0.0016 (0.0400)
sent_leniency_75	0.0913*** (0.0319)	0.0832*** (0.0279)	0.1280*** (0.0357)	0.1190*** (0.0346)
sent_leniency_100	0.0936*** (0.0340)	0.0850*** (0.0289)	0.0857** (0.0369)	0.0769** (0.0348)
Fixed effects	Yes	yes	yes	yes
Additional controls	no	yes	no	yes
p-hat restriction	≥ 0.05	≥ 0.05	none	none
Pre- <i>Blakely</i> mean	0.3099	0.3099	0.4575	0.4575
N	25508	25508	110599	110599

Notes: This table presents results for the regression specified in Equation 6 in the main text. The dependent variable is a binary measure that is equal to one if the minimum sentence imposed equals the maximum amount permitted under the guidelines. Each reported variable represents a quartile of judge leniency, with the most lenient quartile being the omitted group. The sample is limited to 36 months before and after the timing of *Blakely* for consistency with other analyses. Columns 1 and 2 perform the analysis for the $p\text{-hat} \geq 0.05$ restricted sample while columns 3 and 4 contain all observations. All specifications include grid and year fixed effects. The additional controls include age, a male dummy, race, defense type, criminal history points, the type of crime, and annual arrest and population rates. Standard errors are clustered at the district level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.13 Judicial Heterogeneity for Low Probability Cases

	(1) enhanced	(2) enhanced	(3) sentence length	(4) sentence length
<i>Blakely</i>	-0.0134*** (0.0038)	-0.0080** (0.0031)	-0.0344 (0.3100)	0.1650 (0.2720)
DD_50	0.0018 (0.0049)	-0.0015 (0.0046)	-0.3240 (0.2820)	-0.3130 (0.2640)
DD_75	-0.0009 (0.0040)	-0.0032 (0.0036)	-0.4710 (0.3080)	-0.3000 (0.2770)
DD_100	-0.0018 (0.0051)	-0.0060 (0.0050)	-0.5320* (0.304)	-0.3840 (0.272)
p-hat restriction	< 0.05	< 0.03	< 0.05	< 0.03
N	91408	70905	90842	70400

Notes: The following estimates give regression discontinuity results for cases with low propensity of enhancement. Samples in columns 1 and 3 include cases with a propensity score less than 0.05 and columns 2 and 4 include those with a score less than 0.03. All specifications control for defendant characteristics, attorney type, and grid and year fixed effects. Standard errors are clustered at the month level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.14 Heterogeneous Effects of *Blakely* by Other Judge Characteristics

	(1) male	(2) white	(3) tenure	(4) democrat
<i>Panel A: Enhancement Effects</i>				
DD	-0.0139 (0.0161)	-0.0095 (0.0137)	-0.0063 (0.0074)	0.0179 (0.0424)
<i>Panel B: Sentence Length Effects</i>				
DD	0.2597 (1.313)	-0.7187 (1.096)	-1.028* (0.6027)	1.530 (1.971)
N	13227	13077	26692	4803

Notes: For all regressions in Panel A, the dependent variable is a binary for enhanced sentence while Panel B gives results for minimum sentence length. The column headers represent the judge characteristic being tested. DD is an interaction between this characteristic and the timing of *Blakely*. Note that tenure is a binary measure separating the top half tenured vs bottom half tenure judges. All regressions are based on the 0.05 propensity score reduction and include grid and year fixed effects. Standard errors are presented in parentheses and are clustered at the district level.

* p < 0.1, ** p < 0.05, *** p < 0.01

Table B.15 Sample Restrictions and the *Blakely* Discontinuity

	(1) Meaningful Outcomes	(2) Earliest Charges	(3) Meaningful Charges	(4) Felony Charges	(5) Non-Consolidated Charges	(6) Most Severe Charge 1	(7) Most Severe Charge 2
<i>Blakely</i>	0.00130 (0.00360)	-0.00229 (0.00193)	0.00657 (0.00466)	-0.00156 (0.00351)	0.00898 (0.00936)	-0.00117 (0.00434)	-0.00593 (0.00933)
N	332435	249999	245940	227714	193190	127942	96498

Notes: The table shows the sample balance under each sample restriction criterion outlined in Appendix D. From Column (1) to (7), the outcome variables are dummies indicating meaningful outcome, earliest charge, meaningful charge, felony, non-consolidated charge, case aggregation 1, and case aggregation 2. The regression specification in each column is local linear regression with county fixed effects, charging offense fixed effect, and county level controls (population, arrest rates, and employment rates). We restrict the sample to 12 months before and after *Blakely*. Significance level * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix C: A Simple Model of Judicial Sentencing Under Reduced Information

We model a judge who has a preferred sentence length, s^* , which is now dependent on the severity-type of defendant i . The defendant may be one of two unobserved types: $\theta = \{L, H\}$, where L denotes low severity and H high severity. Judges do not directly observe a defendant's severity type θ , but instead infer it from observable case characteristics and prosecutorial actions. Judges have a prior belief about the probability of type: $\Pr(\theta = H) = \pi$. Based on this belief, judges choose a sentence length $s \in [\underline{s}, \bar{s}]$ to maximize the following loss function:

$$U_j(s, \theta) = -(s - s^*(\theta))^2.$$

Note that $s^*(L) < s^*(H)$ for all judges. However, judges may differ in preferences for overall sentence length levels, such that for judges m and n even if θ was known to be H , $s_m^*(H)$ does not necessarily equal $s_n^*(H)$.

We model the prosecutor's decision to file for enhancement as a noisy but informative signal of defendant severity. Let $e = \{0, 1\}$ be the binary enhancement filing choice of the prosecutor. This noisy signal introduces two more probability measures: the probability of enhancement occurring given the defendant type:

$$\Pr(e = 1 | \theta = H) = q_H,$$

$$\Pr(e = 1 | \theta = L) = q_L$$

$$\text{where } q_H > q_L.$$

Utilizing the enhancement signal, the judge arrives at the following posterior probabilities of defendant type:

$$\Pr(H | e = 1) = \frac{q_H \pi}{q_H \pi + q_L (1 - \pi)},$$

$$\Pr(H | e = 0) = \frac{(1 - q_H) \pi}{(1 - q_H) \pi + (1 - q_L) (1 - \pi)}$$

Based on the observed signal e , the judge chooses a sentence equal to the posterior optimal sentence length:

$$s(e) = E[s^*(\theta)|e].$$

When q_H and q_L are sufficiently large, this implies enhancement filings are sufficiently informative. When this occurs, posterior beliefs will diverge and judges will select sentences at or near $s^*(H)$ or $s^*(L)$, depending on the signal type. When enhancement filings become weaker, posterior beliefs converge toward the prior π , and judges increasingly impose pooled sentences near $s^{pool} = \pi s^*(H) + (1 - \pi)s^*(L)$.

Blakely does exactly this - it makes filing for enhancement more costly to the prosecutor, decreasing the probability of enhancement filing such that $q_L^{post} < q_L^{pre}$ and $q_H^{post} < q_H^{pre}$. As these probabilities decrease, the signal weakens and the posterior probabilities of defendant type collapse towards π . This will induce judges to use the pooling equilibrium more frequently post-*Blakely*. This increase in pooling equilibrium leads to a compressing of sentence lengths toward the center of the guideline window.

Appendix D: More Institutional Details

D.1: Sentencing Guidelines

**** Effective for Offenses Committed on or after 12/1/95 Through 11/30/09 ****

OFFENSE CLASS

FELONY PUNISHMENT CHART							
PRIOR RECORD LEVEL							
	I 0 Points	II 1-4 Points	III 5-8 Points	IV 9-14 Points	V 15-18 Points	VI 19+ Points	
A	Death or Life Without Parole						
B1	A	A	A	A	A <i>Life Without Parole</i>	A <i>Life Without Parole</i>	DISPOSITION <i>Aggravated Range</i>
	240 - 300	288 - 360	336 - 420	384 - 480			
	192 - 240	230 - 288	269 - 336	307 - 384	346 - 433	384 - 480	PRESUMPTIVE RANGE
	144 - 192	173 - 230	202 - 269	230 - 307	260 - 346	288 - 384	Mitigated Range
B2	A	A	A	A	A	A	
	157 - 196	189 - 237	220 - 276	251 - 313	282 - 353	313 - 392	
	125 - 157	151 - 189	176 - 220	201 - 251	225 - 282	251 - 313	
	94 - 125	114 - 151	132 - 176	151 - 201	169 - 225	188 - 251	
C	A	A	A	A	A	A	
	73 - 92	100 - 125	116 - 145	133 - 167	151 - 188	168 - 210	
	58 - 73	80 - 100	93 - 116	107 - 133	121 - 151	135 - 168	
	44 - 58	60 - 80	70 - 93	80 - 107	90 - 121	101 - 135	
D	A	A	A	A	A	A	
	64 - 80	77 - 95	103 - 129	117 - 146	133 - 167	146 - 183	
	51 - 64	61 - 77	82 - 103	94 - 117	107 - 133	117 - 146	
	38 - 51	46 - 61	61 - 82	71 - 94	80 - 107	88 - 117	
E	I/A	I/A	A	A	A	A	
	25 - 31	29 - 36	34 - 42	46 - 58	53 - 66	59 - 74	
	20 - 25	23 - 29	27 - 34	37 - 46	42 - 53	47 - 59	
	15 - 20	17 - 23	20 - 27	28 - 37	32 - 42	35 - 47	
F	I/A	I/A	I/A	A	A	A	
	16 - 20	19 - 24	21 - 26	25 - 31	34 - 42	39 - 49	
	13 - 16	15 - 19	17 - 21	20 - 25	27 - 34	31 - 39	
	10 - 13	11 - 15	13 - 17	15 - 20	20 - 27	23 - 31	
G	I/A	I/A	I/A	I/A	A	A	
	13 - 16	15 - 19	16 - 20	20 - 25	21 - 26	29 - 36	
	10 - 13	12 - 15	13 - 16	16 - 20	17 - 21	23 - 29	
	8 - 10	9 - 12	10 - 13	12 - 16	13 - 17	17 - 23	
H	C/I/A	I/A	I/A	I/A	I/A	A	
	6 - 8	8 - 10	10 - 12	11 - 14	15 - 19	20 - 25	
	5 - 6	6 - 8	8 - 10	9 - 11	12 - 15	16 - 20	
	4 - 5	4 - 6	6 - 8	7 - 9	9 - 12	12 - 16	
I	C	C/I	I	I/A	I/A	I/A	
	6 - 8	6 - 8	6 - 8	8 - 10	9 - 11	10 - 12	
	4 - 6	4 - 6	5 - 6	6 - 8	7 - 9	8 - 10	
	3 - 4	3 - 4	4 - 5	4 - 6	5 - 7	6 - 8	

A – Active Punishment I – Intermediate Punishment C – Community Punishment

Numbers shown are in months and represent the range of minimum sentences

A - Active Punishment I - Intermediate Punishment C - Community Punishment
Numbers shown are in months and represent the range of minimum sentences

D.2 Statutory Aggravating Factors

Aggravating factors listed in § 15A-1340.16.(d):

- (1) The defendant induced others to participate in the commission of the offense or occupied a position of leadership or dominance of other participants.
- (2) The defendant joined with more than one other person in committing the offense and was not charged with committing a conspiracy.
 - (2a) The offense was committed for the benefit of, or at the direction of, any criminal gang as defined by G.S. 14-50.16A(1), with the specific intent to promote, further, or assist in any criminal conduct by gang members, and the defendant was not charged with committing a conspiracy.
- (3) The offense was committed for the purpose of avoiding or preventing a lawful arrest or effecting an escape from custody.
- (4) The defendant was hired or paid to commit the offense.
- (5) The offense was committed to disrupt or hinder the lawful exercise of any governmental function or the enforcement of laws.
- (6) The offense was committed against or proximately caused serious injury to a present or former law enforcement officer, employee of the Division of Adult Correction and Juvenile Justice of the Department of Public Safety, jailer, fireman, emergency medical technician, ambulance attendant, social worker, justice or judge, clerk or assistant or deputy clerk of court, magistrate, prosecutor, juror, or witness against the defendant, while engaged in the performance of that person's official duties or because of the exercise of that person's official duties.
 - (6a) The offense was committed against or proximately caused serious harm as defined in G.S. 14-163.1 or death to a law enforcement agency animal, an assistance animal, or a search and rescue animal as defined in G.S. 14-163.1, while engaged in the performance of the animal's official duties.
- (7) The offense was especially heinous, atrocious, or cruel.
- (8) The defendant knowingly created a great risk of death to more than one person by means of a weapon or device which would normally be hazardous to the lives of more than one person.
- (9) The defendant held public elected or appointed office or public employment at the time of the offense and the offense directly related to the conduct of the office or employment.
 - (9a) The defendant is a firefighter or rescue squad worker, and the offense is directly related to service as a firefighter or rescue squad worker.
- (10) The defendant was armed with or used a deadly weapon at the time of the crime.
- (11) The victim was very young, or very old, or mentally or physically infirm, or handicapped.

- (12) The defendant committed the offense while on pretrial release on another charge.
 - (12a) The defendant has, during the 10-year period prior to the commission of the offense for which the defendant is being sentenced, been found by a court of this State to be in willful violation of the conditions of probation imposed pursuant to a suspended sentence or been found by the Post-Release Supervision and Parole Commission to be in willful violation of a condition of parole or post-release supervision imposed pursuant to release from incarceration.
- (13) The defendant involved a person under the age of 16 in the commission of the crime.
 - (13a) The defendant committed an offense and knew or reasonably should have known that a person under the age of 18 who was not involved in the commission of the offense was in a position to see or hear the offense.
- (14) The offense involved an attempted or actual taking of property of great monetary value or damage causing great monetary loss, or the offense involved an unusually large quantity of contraband.
- (15) The defendant took advantage of a position of trust or confidence, including a domestic relationship, to commit the offense.
- (16) The offense involved the sale or delivery of a controlled substance to a minor.
 - (16a) The offense is the manufacture of methamphetamine and was committed where a person under the age of 18 lives, was present, or was otherwise endangered by exposure to the drug, its ingredients, its by-products, or its waste.
 - (16b) The offense is the manufacture of methamphetamine and was committed in a dwelling that is one of four or more contiguous dwellings.
- (17) The offense for which the defendant stands convicted was committed against a victim because of the victim's race, color, religion, nationality, or country of origin.
- (18) The defendant does not support the defendant's family.
 - (18a) The defendant has previously been adjudicated delinquent for an offense that would be a Class A, B1, B2, C, D or E felony if committed by an adult.
- (19) The serious injury inflicted upon the victim is permanent and debilitating.
- (20) Any other aggravating factor reasonably related to the purposes of sentencing.

Appendix E: Data cleaning

Our main dataset comes from the North Carolina Administrative Office of Courts. The data cleaning is largely based on Silveria (2017) which used the same dataset.

The initial database includes all of criminal charges filed in North Carolina. The state court system typically defines “case” with a common file number, which is frequently associated with multiple counts of charges. However, such definition of “case” does not accurately reflect the reality of criminal cases are resolved and could cause many issues in empirical analysis. First, the same file number reappears in the data if defendants violate probation conditions, which should not be counted as new criminal offenses. Second, the data suggests that defendants are often disposed of various charges associated with different file numbers on the same date. What’s more, anecdotal evidence suggests that plea bargain applies to all outstanding charges. As a result, we decided to adopt an alternative definition of “case” by using a pair of individual identifier and disposition date, which was used in Abrams & Fackler (2020).

The data cleaning begins with a total number of 2,364,028 charges which were filed in the North Carolina superior courts between 1995 and 2009. This paper focuses on the sentence enhancement outcomes among defendants charged with felony offenses. In North Carolina, superior courts have the exclusive jurisdiction over all felony cases. We dropped all the cases and charges resolved at the NC district courts because the district courts only hear misdemeanor and infraction cases which are not relevant to our analysis of sentence enhancements.

Firstly, we keep charges with meaningful outcomes, meaning either dismissed or determined by judges or juries. This procedure eliminated irrelevant cases where there are changes of venues, decisions were made state district courts, etc. The observations dropped to 1,774,336 (25% decrease). Secondly, among charges associated with the same file numbers, we keep the counts which were disposed at the earliest date, which eliminated all the charges filed due to probation violations. The number of observations dropped to 1,745,225 (1.6% decrease). Thirdly, we dropped all charges which do not consist of meaningful information on charged offense and case outcomes. The observations dropped to 1,621,345 (7.1% decrease). Fourthly, we drop the charges where defendants are charged of misdemeanor offenses as the sentencing guidelines in Figure D.1 does not apply to misdemeanor offenses. This step reduces the observation to 1,379,454 (14.9% decrease). Lastly, we dropped charges which are consolidated with other ones, which further reduced the number of observations to 937,817 (32% decrease).¹ North Carolina state laws allow courts to consolidate less serious charges and to only issue one judgment on the basis of the most serious offense. Consolidated charges do not have influence the final sentence outcomes.

Then, we further aggregate charges at two levels in order to define criminal “cases.” First, for each file number, we keep the count associated with the most severe charge.² This step

¹ The data includes a variable called “crdcfjf” and “crdcfjo,” which identifies the charge being consolidated to.

² For charges associated with the same file number, we first order them on the basis of punishment severity, which we measure by using a pair of punishment type and minimum sentence length. We keep the count(s) associated the most severe punishment. However, such procedure could not help us determine the most severe charge in some

reduces the number of observations to 713,404 (24% decrease). Furthermore, we implemented the same aggregation procedure for each pair of individual identifiers and disposition date. The number of observations dropped to 504,498 (27.5% decrease). In the end, we are able to proxy criminal “case” by the most severe charge that defendants are disposed on the same date. In developing the empirical analysis, we further restrict time frame from 2001 to 2007.

One might be concerned that our extensive data cleaning process might introduce sample selection bias that might be correlated with the *Blakely* discontinuity. To mitigate such concern, we performed both graphical and regression analysis showing that the fraction of the sample that survives each sample restriction criteria is smooth at the discontinuity. Figure A.9 shows that the fraction of sample, calculated at the monthly level, that survived each of the seven data cleaning criteria. We implemented density test for each sample restriction criterion around the timing of the discontinuity. The graphs that none of the sample restriction criterion creates artificial discontinuity at the timing of *Blakely*. In addition, we performed regression analysis examining whether *Blakely* affects the likelihood of our data surviving each sample restriction criterion. Table B.10 shows the results. All the regression coefficients are small and statistically indistinguishable from 0. The regression results reassure that none of our sample restrictions criteria introduced artificial discontinuities at the timing of *Blakely*.

situations. For example, dismissed cases or charges with identical sentences. For those cases, we keep the charge associated with the most severe filed offense.